Project Proposal: Graduate Independent Study in Computer Science

Project Title: Vector Database Comparison for Semantic Search Performance

Student Name: Shivali Mate

Faculty Mentor: Professor Feng Chen

Course: Graduate Independent Study (Computer Science)

Credit Hours: 3

Semester: Spring 2026

1. Work Plan

This independent study will investigate and compare the capabilities of leading open-source and cloud based vector databases for semantic search. The project will include benchmarking key performance indicators like retrieval accuracy, speed, and scalability, alongside the development of a simple semantic search tool using Python and modern embedding models. The outcome is an analytical report and demonstration prototype outlining comparative strengths and weaknesses.

2. Expected Student Work Hours

The student will dedicate 9-10 hours per week to the study. This time includes reading current literature, database setup and experimentation, coding, documentation, and meetings with the faculty mentor. This aligns with the standards for a 3 credit graduate independent study.

3. Required Meetings with Mentor and Team

- Regular Weekly Meeting: One meeting per week with the faculty mentor, scheduled for progress updates, troubleshooting, and milestone reviews.
- Ad-hoc consultations as needed for technical or scope adjustments.

4. Required Readings and Assignments

As a research based study, assignments will be milestone driven rather than a fixed set of tasks.

Readings:
Core readings will be selected and reviewed by the student, focusing on vector databases, semantic search, and benchmarking methodology. Literature review will be ongoing, drawing from academic and industry sources relevant to modern database research.

Assignments & Milestones:
The project will be broken down into phases. Specific tasks and deadlines will be determined collaboratively with the mentor and reviewed during weekly meetings.

5. Assessment and Grading Criteria

The final grade will be determined based on the student's consistent engagement, the quality of their contributions, and their ability to document and communicate their work effectively.

- **Weekly Progress and Participation:** Assessed based on the student's clear and prepared updates during weekly group meetings, active participation in discussions, and consistent progress toward agreed upon milestones.
- **Technical Contribution & Execution:** Evaluated on the quality, functionality, and rigor of the code, models, and analyses produced. This includes the successful implementation of assigned tasks.
- **Documentation and Final Reporting:** Graded on the clarity, completeness, and quality of technical documentation, including code comments, slides for presentations, and a final summary report of the project work and results